Applied integral test to the hyper-harmonic series.

$$\sum_{n=1}^{\infty} \frac{1}{n^a}$$

we found

- converges for $a > 1$
- diverges for $0 \leq a \leq 1$

Comparison tests

The direct or limit comparison test work by comparing the series in question to a series with a known convergence status and then can state the status of the series in question.

Theorem 2.3.7.

We're examining $\sum_{n=1}^{\infty} a_n$, and we have $\sum_{n=1}^{\infty} c_n$ which is a known convergent series and $\sum_{n=1}^{\infty} d_n$ a known divergent series. Assume all terms a_m, c_m, d_m are non-negative.

- if for all $m \geq 1$ $a_m \leq c_m$ then $\sum_{m=1}^{\infty} a_m$ also converges.
- if for all $m \geq 1$ $d_m \leq a_m$ then $\sum_{m=1}^{\infty} a_m$ also diverges

Application example.

Consider the series $\sum_{n=1}^{\infty} \underset{\swarrow a_n}{\frac{1}{3^n + n^2}}$.

We notice that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a known convergent hyper-harmonic series.

Also, $\sum_{n=1}^{\infty} \frac{1}{3^n}$ is a known convergent geometric series. We could take either of these as the comparison series $\sum_{m=1}^{\infty} c_m$. Lets choose $c_n = \frac{1}{n^2}$

$$a_n = \boxed{\frac{1}{3^n+n^2} < \frac{1}{n^2}}$$

So by the comparison test

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{3^n+n^2} \text{ also converges.}}$$

Or summing up the limit comparison test, more vaguely,

If $\displaystyle\lim_{n\to\infty} \frac{a_n}{b_n} = L$.

then for n big enough.

$$\frac{a_n}{b_n} \approx L$$

$$\Rightarrow \quad a_n \approx L\, b_n$$

$$\Rightarrow \quad \sum_{n=N}^{\infty} a_n \approx L \sum_{n=N}^{\infty} b_n \, .$$

and so $\sum a_n$ converges iff

$\sum b_n$ converges.

Example

Consider $\sum_{n=1}^{\infty} a_n$,

$$a_n = \frac{\sqrt{n} + 3}{n^2 - n + 1}$$

How to choose the comparison series?

Consider the dominant behaviour in numerator and denominator for large n

$$a_n \sim \frac{\sqrt{n}}{n^2} = n^{-3/2} = \frac{1}{n^{3/2}}$$

So choose $c_n = \frac{1}{n^{3/2}}$

and notice $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ is a known convergent hyper-harmonic series.

Consider.

$$\boxed{\lim_{n\to\infty} \frac{a_n}{c_n}} = \lim_{n\to\infty} \frac{\sqrt{n}}{n^2-n+1} \cdot n^{3/2}$$

$$= \lim_{n\to\infty} \frac{n^2}{n^2-n+1}$$

$$= \lim_{n\to\infty} \frac{1}{1-1/n+1/n^2}, \quad \text{div. above/below by } n^2$$

$$\boxed{= 1,}$$ by algebra of limits theorem for sequences and $\frac{1}{n}, \frac{1}{n^2} \to 0$ as $n \to \infty$.

So from ~~the~~ Limit Comparison Test we get $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2-n+1}$ also converges.

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Ratio - test

Consider $\sum_{m=1}^{\infty} a_m$, with $a_m > 0$

and suppose $\lim_{m \to \infty} \frac{a_{m+1}}{a_m} = L$

- If $0 \le L < 1$ then $\sum_{m=1}^{\infty} a_m$ converges.
- If $L > 1$ then $\sum_{m=1}^{\infty} a_m$ diverges
- $L = 1$, ~~no~~ ratio-test gives no conclusion.

<u>Examples of application of ratio test.</u>

Consider. $\sum_{n=1}^{\infty} \frac{2}{n!}$ $\qquad a_m = \frac{2}{m!}$

$2/n! \longrightarrow 0$ as $n \to \infty$ pretty fast

so probably a convergent series.

Consider

$$\lim_{m \to \infty} \frac{a_{m+1}}{a_m} = \lim_{m \to \infty} \frac{2/(m+1)!}{2/m!}$$

$$= \lim_{m \to \infty} \left(\frac{1}{m+1} \right) = 0 = L$$

Since $0 \le L < 1$ so by the ratio-test $\sum_{n=1}^{\infty} 2/n!$ is a convergent series.

Another ex

Consider $\displaystyle\sum_{n=1}^{\infty} \frac{3^n}{n^3}$

so $a_n = 3^n/n^3$

$$\boxed{\lim_{n\to\infty} \frac{a_{n+1}}{a_n}} = \lim_{n\to\infty} \frac{3^{n+1}}{(n+1)^3} \frac{n^3}{3^n}$$

$$= 3 \lim_{n\to\infty} \frac{n^3}{(n+1)^3}$$

$$= 3 \lim_{n\to\infty} \left(\frac{n^3}{n^3+3n^2+3n+1} \right)$$

$$= 3 \lim_{n\to\infty} \left(\frac{1}{1 + 3/n + 3/n^2 + 1/n^3} \right)$$

$\boxed{= 3}$, by the algebra of limits theorem and $1/n, 1/n^2, 1/n^3 \to 0$ as $n \to \infty$.

$L = 3$

$L = 3 > 1$ so ratio test tells us that $\sum_{n=1}^{\infty} \frac{3^n}{n^3}$ is divergent.

Notice the General Term Test also tells us this is divergent.

since $\frac{3^n}{n^3}$ is increasing and unbounded

An example of the $L=1$ inconclusive case is provided by the harmonic series. $\sum_{n=1}^{\infty} \frac{1}{n}$ $\qquad a_n = \frac{1}{n}$

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{n}{n+1}$$

$$= \lim_{n\to\infty} \frac{1}{1+1/n}$$

$$= 1 = L$$

Continue your reading on Root test and Alternating Series test.

<u>Telescoping series</u> A class of series where a <u>lot</u> of cancellation happens in the partial sums allowing us to determine the sum of the series. Uses partial fraction expansions of rational function.

<u>Example</u>

Q14 from tutorial ~~set~~ sheet.

Consider $\sum_{n=1}^{\infty} b_n$

where $b_n = \dfrac{8}{n^2+6n+5}$.

$$= \frac{8}{(n+5)(n+1)}$$

(from the partial fractions technique.)

$$= \frac{\alpha}{n+5} + \frac{\beta}{n+1},$$ for some pair α, β.

$$= \frac{\alpha(n+1) + \beta(n+5)}{(n+5)(n+1)}$$

$$= \frac{(\alpha+\beta)n + (\alpha+5\beta)}{(n+5)(n+1)}$$

Comparing this back to $\dfrac{8}{(n+5)(n+1)}$

So for all $n \geq 1$ $\quad 8 = (\alpha+\beta)n + (\alpha+5\beta)$

$$\Rightarrow \quad \left\{ \begin{array}{l} 0 = \alpha+\beta \\ 8 = \alpha+5\beta \end{array} \right\}$$

$$\Rightarrow \quad \left\{ \begin{array}{l} \alpha = -\beta \\ 8 = 4\beta \end{array} \right\}.$$

$$\Rightarrow \quad \beta = 2, \quad \alpha = -2.$$

So $\quad b_n = \dfrac{-2}{n+5} + \dfrac{2}{n+1}.$

Remember

$$\sum_{n=1}^{\infty} b_n = \lim_{k\to\infty}\left(\sum_{n=1}^{k} b_n\right), \text{ def of } \sum_{1}^{\infty}.$$

$$\sum_{n=1}^{k} b_n = \sum_{n=1}^{k}\left(\frac{-2}{n+5}+\frac{2}{n+1}\right)$$

$$= \left(\cancel{\frac{-2}{6}}+\frac{2}{2}\right)+\left(\cancel{\frac{-2}{7}}+\frac{2}{3}\right)$$

$$+\left(\cancel{\frac{-2}{8}}+\frac{2}{4}\right)+\left(\cancel{\frac{-2}{9}}+\frac{2}{5}\right)$$

$$+\left(-\frac{2}{10}+\cancel{\frac{2}{6}}\right)+\left(\frac{-2}{11}+\cancel{\frac{2}{7}}\right)+\cdots+$$

$$+\left(\frac{-2}{k+4}+\frac{2}{k}\right)+\left(\frac{-2}{k+5}+\frac{2}{k+1}\right)$$

Notice that almost every negative cancels with a later positive, so all that remains will be the early positive and some later negative

$$\sum_{n=1}^{k} b_n = \frac{2}{2}+\frac{2}{3}+\frac{2}{4}+\frac{2}{5}$$

$$- \frac{2}{k+2} - \frac{2}{k+3} - \frac{2}{k+4} - \frac{2}{k+5}.$$

So $\boxed{\sum_{n=1}^{\infty} b_n} = \lim_{k \to \infty} \sum_{n=1}^{k} b_n$

$$= \lim_{k \to \infty} \left(\frac{2}{2} + \frac{2}{3} + \frac{2}{4} + \frac{2}{5} - \frac{2}{k+2} - \frac{2}{k+3} - \frac{2}{k+4} - \frac{2}{k+5} \right)$$

$= \frac{2}{2} + \frac{2}{3} + \frac{2}{4} + \frac{2}{5}$, by algebra of limits theorem and $\frac{1}{k+i} \to 0$ as $k \to \infty$ for $i = 2, \dots, 5$.

$$= \frac{60 + 40 + 30 + 24}{60}$$

$$= \frac{154}{60} = \boxed{\frac{77}{30}}$$

So the telescoping /cancelling nature of the partial sum has allowed us to determine $\sum_{n=1}^{\infty} b_n$.

See section 6.5 in APEX textbook for full info on partial fraction expansions